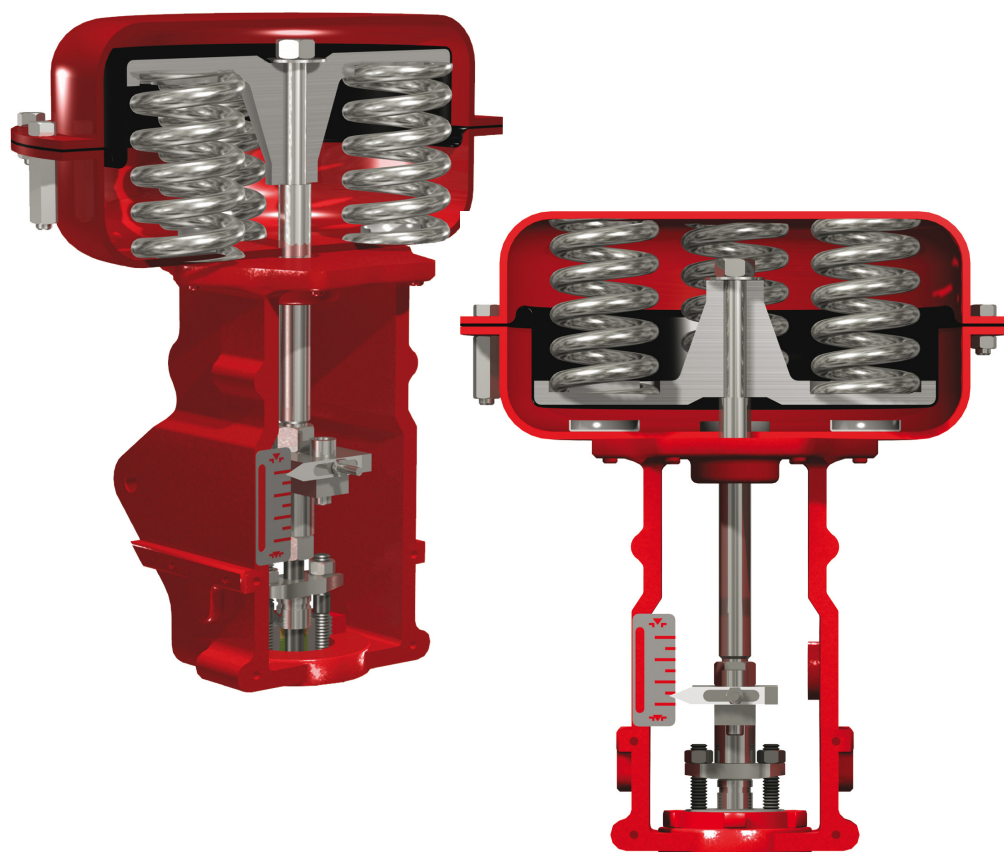
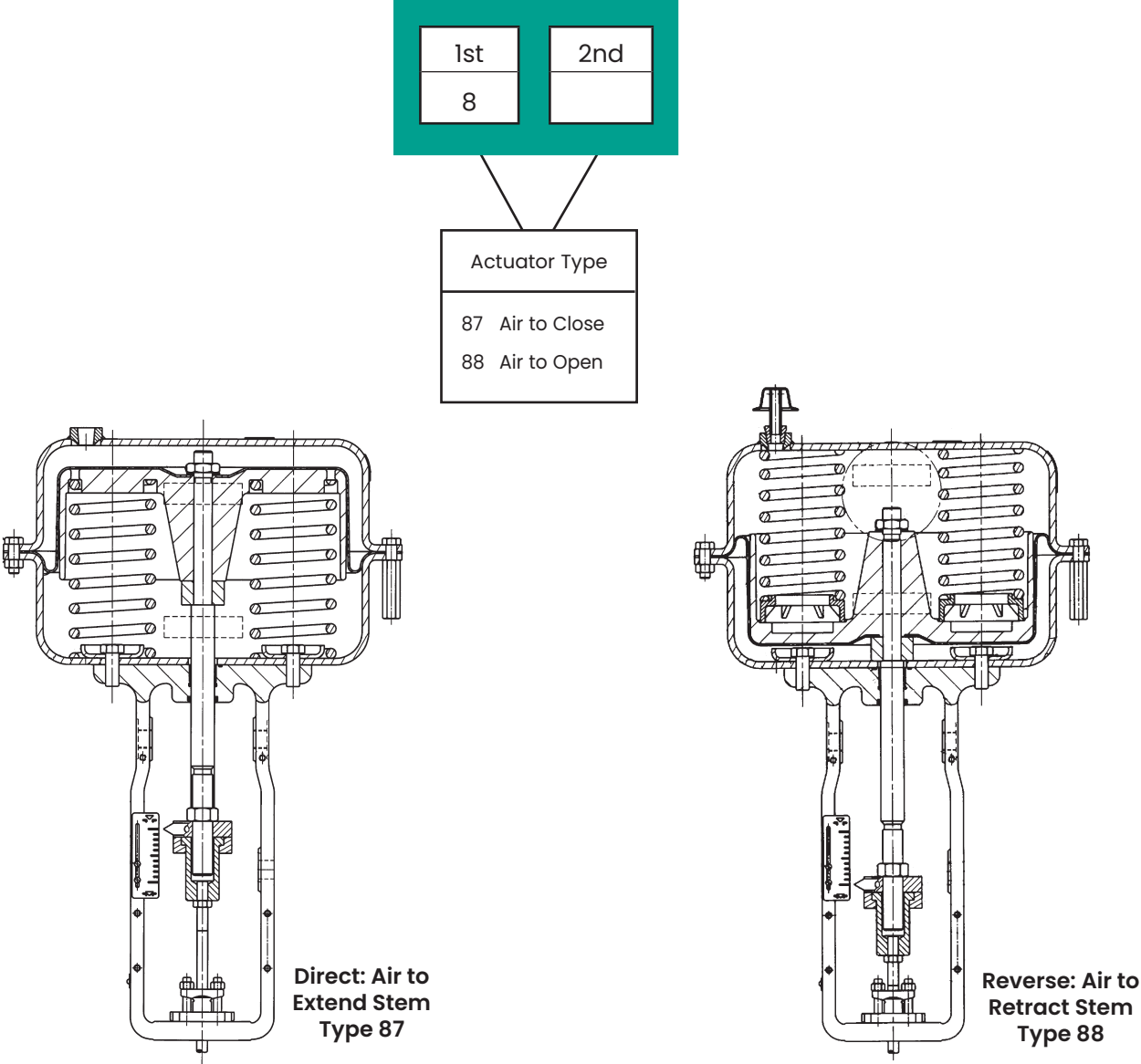


Masoneilan™ 87/88 Series Actuators

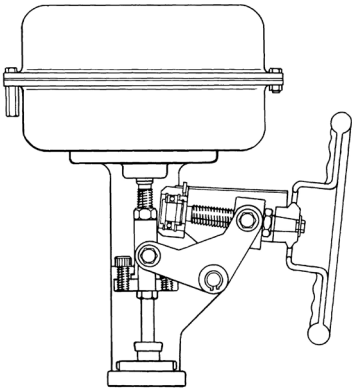
Pneumatic Multi-Spring Diaphragm Actuators



Numbering System



Handwheel



Actuator Size	Maximum Travel		Equivalent Air Supply		Number of Turns
	Inches	mm	psi	bar	
6	0.8	20	60	4.1	5
10	1.5	38	60	4.1	9
16	2.5	63	60	4.1	17
23	2.5	63	60	4.1	25
23L	4.0	102	60	4.1	440

Specifications

Actuator Size	Diaphragm Effective Area		Stem Travel		Max Supply	
	sq. in.	sq. cm.	inches	mm	psi	bar
6	60	387	0.8	20	60	4.1
10	100	645	0.8 / 1.5	20 / 38	60	4.1
16	160	1032	0.8 / 1.5 / 2.0 / 2.5	20 / 38 / 50 / 63	60	4.1
23	230	1484	0.8 / 1.5 / 2.0 / 2.5	20 / 38 / 50 / 63	60	4.1
23L	230	1484	3.0 / 3.5 / 4.0	76 / 89 / 102	60	4.1

Alternate Design Temperature Ranges

Standard Temperature Range				Extended Temperature Range				Ultra-Low Temperature Range			
Min Temp		Max Temp		Min Temp		Max Temp		Min Temp		Max Temp	
°C	°F	°C	°F	°C	°F	°C	°F	°C	°F	°C	°F
-30	-22	83	181.4	-40	-40	100	212	-50	-58	100	212

Note: See Materials Table for Alternative Substitutions

Standard Springs

Actuator Size 6, 10, 16, and 23

Spring Type	Spring Qty	Placement
3-15	3	Lower Level
6-30	6	Lower Level
11-23	3	Upper Level
21-45	6	Upper Level

Actuator Size 23L

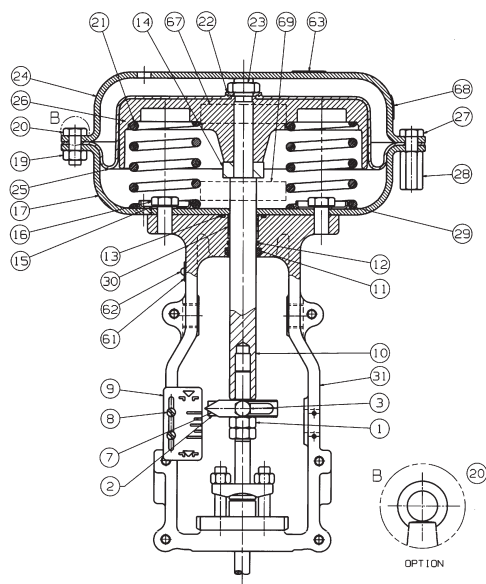
Actuator Model	Stem Travel		Spring Type	Spring Qty	Placement
	Inches	mm			
87	4	102	3-15	3	Lower Level
			6-30	6	Lower Level
			11-23	3	Upper Level
			21-45	6	Upper Level
	3.5	89	6-27	6	Lower Level
88	3	76	6-24	6	Lower Level
	4	102	3-15	3	Lower Level
			6-30	6	Lower Level
			11-23	3	Upper Level
			21-45	6	Upper Level
	3.5	89	9-30	6	Lower Level
			24-45	6	Upper Level
	3	76	12-30	6	Lower Level
			27-45	6	Upper Level

Air Connections

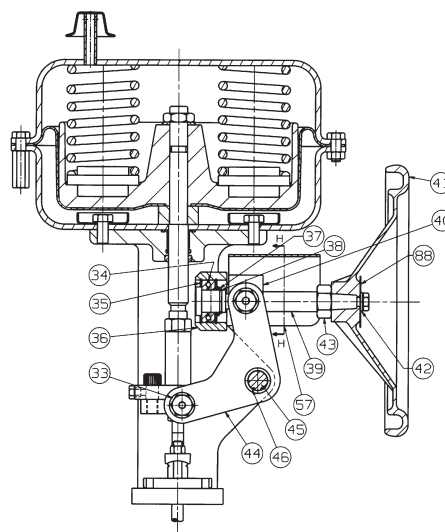
Standard: 1/4" NPT - Size 6, 10 1/2" NPT - Size 16, 23

Optional: 1/2" NPT - Size 6, 10

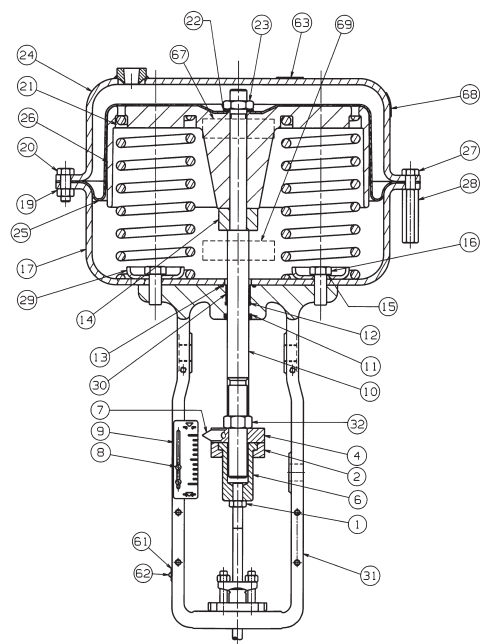
Materials



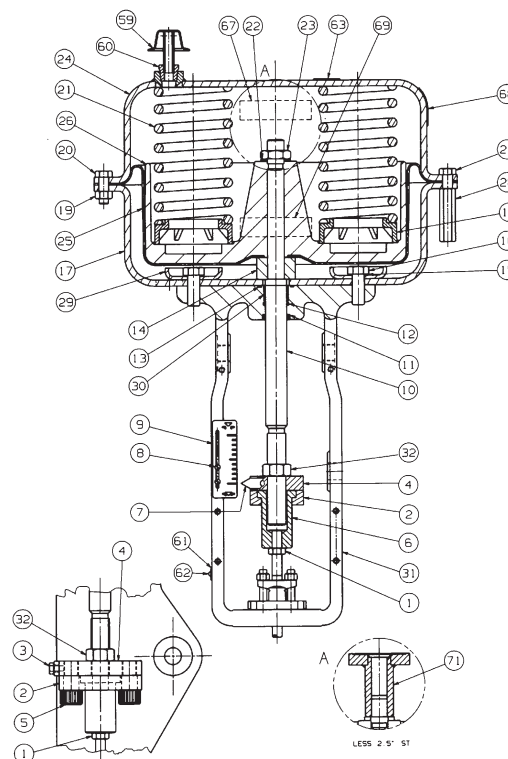
Size 6 Actuator



**Size 10 Actuator
with Optional Handwheel**

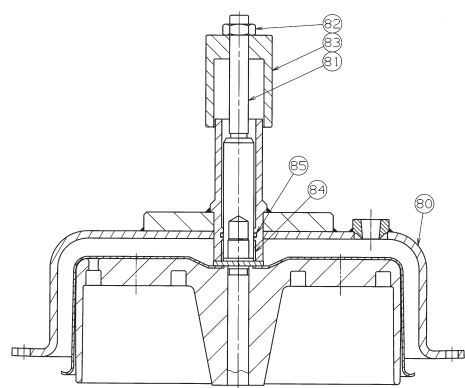


**Size 16 Actuator Air to Close
(Model 87)**

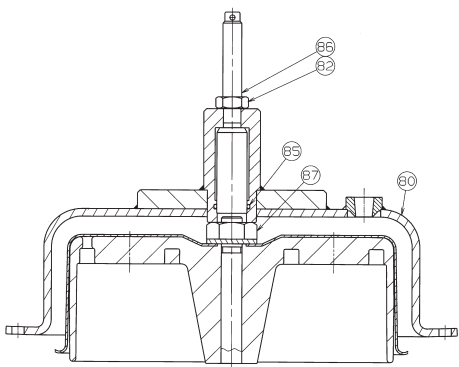


**Size 16 Actuator Air to Open
(Model 88)**

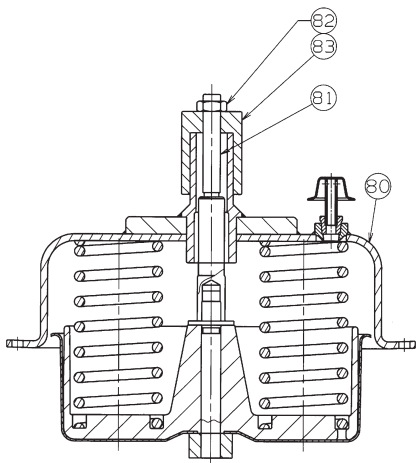
Materials



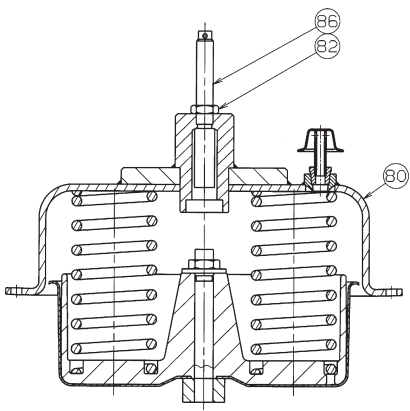
Model 87 - Down Stop
Optional Adjustable Limit Stop



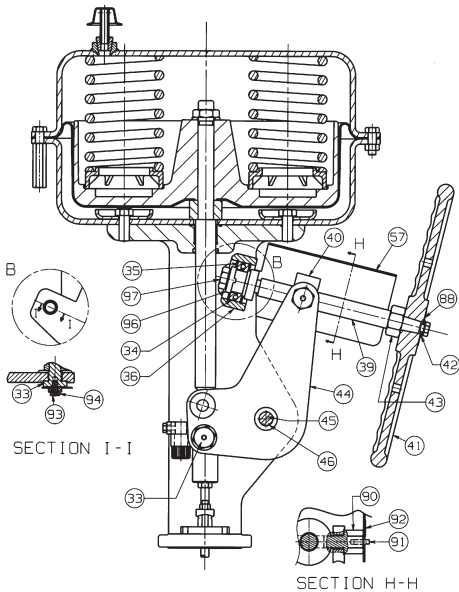
Model 87 - Up Stop
Optional Adjustable Limit Stop



Model 88 - Down Stop
Optional Adjustable Limit Stop



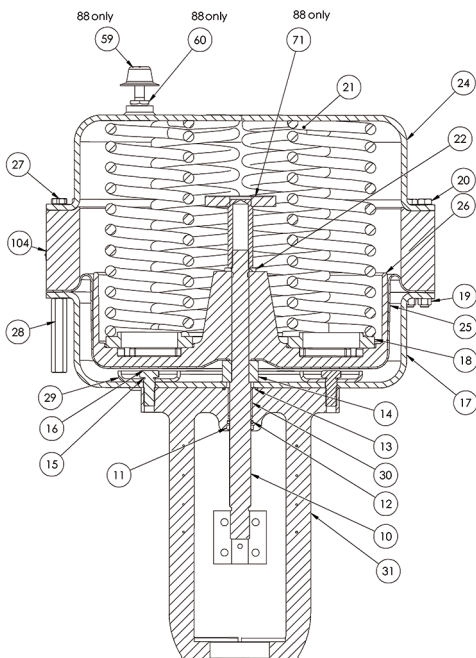
Model 88 - Up Stop
Optional Adjustable Limit Stop



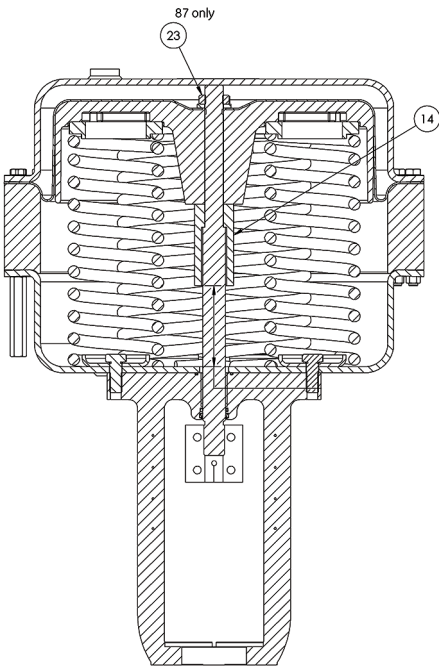
Size 23 Actuator with Optional Handwheel

Materials

Size 23L Without Optional Handwheel

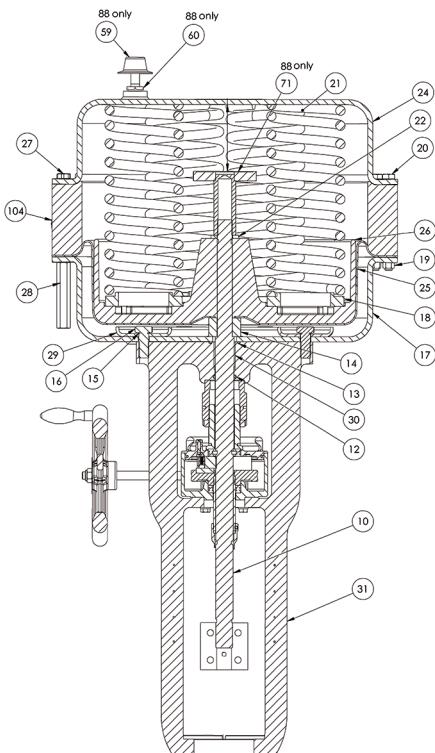


Model 88

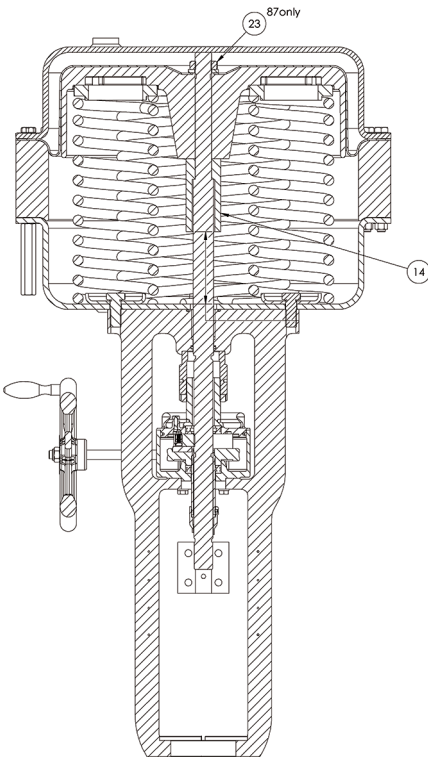


Model 87

With Optional Handwheel



Model 88



Model 87

Materials

Parts Reference for Actuator Sizes 6, 10, 16, 23 and 23L

Ref No.	Description	Standard Temp Range Material	Extended Temp Range Material ⁽³⁾	Ultra-Low Temp Range Material ⁽³⁾
1	Lock Nut	Stainless Steel	(2)	(2)
2	Stem Connector-Bottom	Carbon Steel, Zinc Plate	(2)	(2)
3	Cap Screw-Pointer	Carbon Steel, Zinc Plate	(2)	(2)
4	Stem Connector-Top	Carbon Steel, Zinc Plate	(2)	(2)
5	Cap Screw-Connector	Carbon Steel, Zinc Plate	(2)	ASTM A193 Cr B8 Cl 2 ⁽¹⁾
6	Connector Insert	Carbon Steel, Zinc Plate	(2)	(2)
7	Pointer	Stainless Steel	(2)	(2)
8	Screw-Pan Head	Carbon Steel, Zinc Plate	(2)	(2)
9	Scale-Travel	Stainless Steel	(2)	(2)
10	Actuator Stem	Stainless Steel	(2)	(2)
10a	Actuator Stem	Stainless Steel	(2)	(2)
11	Stem Wiper	Polyurethane	(2)	(2)
12	O-ring	Buna-N	Silicone ⁽¹⁾	Silicone ⁽¹⁾
13	O-ring	Buna-N	Silicone ⁽¹⁾	Silicone ⁽¹⁾
14	Spacer	Carbon Steel, Zinc Plate	(2)	(2)
15	Seal Washer	Steel & Rubber	(2)	(2)
16	Cap Screw-Yoke	Carbon Steel, Zinc Plate	(2)	(2)
17	Lower Diaphragm Case	Carbon Steel	(2)	(2)
18	Spring Spacer	Carbon Steel	(2)	(2)
19	Hex Nut	Carbon Steel, Zinc Plate	(2)	(2)
20	Cap Screw-Hex Head	Carbon Steel, Zinc Plate	(2)	(2)
21	Spring	Alloy Steel	(2)	(2)
22	Washer Flat	Carbon Steel, Zinc Plate	(2)	(2)
23	Nut-Jam	Carbon Steel, Zinc Plate	(2)	(2)
24	Upper Diaphragm Case	Carbon Steel	(2)	(2)
25	Diaphragm	Buna-N/Polyester	Silicone Rubber with Dacron Fabric ⁽¹⁾	Silicone Rubber with Dacron Fabric ⁽¹⁾
26	Diaphragm Plate	Aluminum Alloys	(2)	(2)
27	Cap Screw-Comp	Carbon Steel	(2)	(2)
28	Compression Nut	Carbon Steel, Zinc Plate	(2)	(2)
29	Spring Guide	Carbon Steel, Zinc Plate	(2)	(2)
30	Bushing	PTFE/Steel	(2)	(2)
31	Yoke	Cast Iron, ASTM A216 Gr WCC ⁽¹⁾	ASTM A352 Gr. LCC HRc 22 Max ⁽¹⁾	ASTM A352 Gr. LC1 HRc 22 Max ⁽¹⁾
32	Lock Nut	Stainless Steel	(2)	(2)
33	Pivot Pin	Carbon Steel, Zinc Plate	(2)	(2)
34	Thrust Bearing	Carbon Steel	(2)	(2)
36	HW Pivot	Carbon Steel, Zinc Plate	(2)	(2)
37	Bearing Ring	Carbon Steel, Zinc Plate	(2)	(2)
38	Retaining Ring	Carbon Steel	(2)	(2)

1. Alternative Materials
2. Same as Standard
3. Refer to Alternative Temperature Range under Specifications on page 3 for full range details.

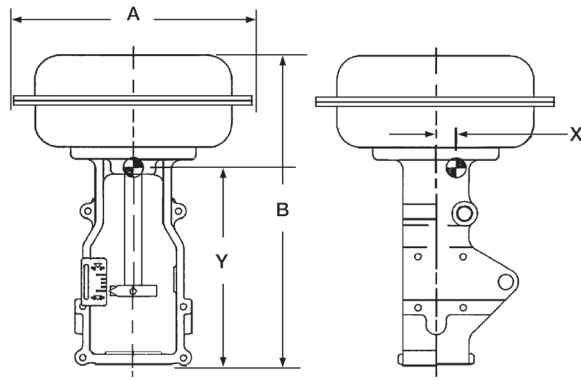
Materials (Cont.)

Parts Reference for Actuator Sizes 6, 10, 16, 23 and 23L

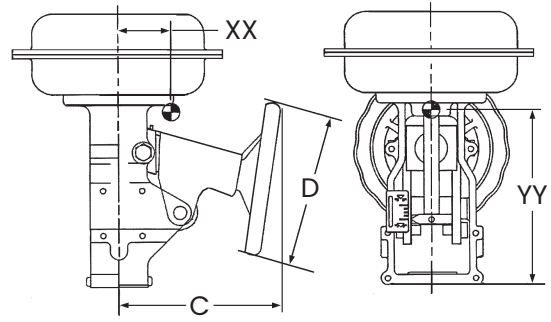
Ref No.	Description	Standard Temp Range Material	Extended Temp Range Material ⁽³⁾	Ultra-Low Temp Range Material ⁽³⁾
39	H/W Stem	Carbon Steel, Zinc Plate	(2)	(2)
40	Traveling Nut	Leaded Steel	(2)	(2)
41	Handwheel	Cast Iron	(2)	(2)
42	Washer-Flat	Carbon Steel, Zinc Plate	(2)	(2)
43	HW Limitation Nut	Carbon Steel, Zinc Plate	(2)	(2)
44	HW Lever	Carbon Steel	(2)	(2)
45	Lever Pin	Carbon Steel, Zinc Plate	(2)	(2)
46	Retaining Ring-Lever Pin	Carbon Steel	(2)	(2)
57	HW Cover	Carbon Steel, Zinc Plate	(2)	(2)
59	Vent Plug	Carbon Steel, Zinc Plate	(2)	(2)
60	Pipe Fitting	Carbon Steel, Zinc Plate	(2)	(2)
63	Information Label	Vinyl	(2)	(2)
67	Warning Label	Vinyl	(2)	(2)
68	Warning Label	Vinyl	(2)	(2)
69	Warning Label	Vinyl	(2)	(2)
71	Travel Stop	Carbon Steel, Zinc Plate	(2)	(2)
80	Upper Diaphragm Case	Carbon Steel	(2)	(2)
81	Stem Extension	Stainless Steel	(2)	(2)
82	Lock Nut	Stainless Steel	(2)	(2)
83	Limit Stop	Stainless Steel	(2)	(2)
84	Bushing	Steel/ PTFE Lined	(2)	(2)
85	O-ring	Buna-N	Silicone ⁽¹⁾	Silicone ⁽¹⁾
86	Stop	Stainless Steel	(2)	(2)
87	Stop	Carbon Steel	(2)	(2)
88	Plate-Rotation Arrow	Stainless Steel	(2)	(2)
90	Pivot Pin-HW Lock	Carbon Steel, Zinc Plate	(2)	(2)
91	Handwheel Lock Pin	Stainless Steel	(2)	(2)
92	Information Plate-HW	Vinyl	(2)	(2)
93	Cap Screw-Hex Head	Carbon Steel	(2)	(2)
94	Washer-Flat	Carbon Steel, Zinc Plate	(2)	(2)
96	End Flange	Carbon Steel, Zinc Plate	(2)	(2)
97	Cap Screw-HW Stem	Carbon Steel, Zinc Plate	(2)	(2)
105	Recessed Head Screw	Carbon Steel	(2)	(2)
106	Indicator Plate	Carbon Steel	(2)	(2)
107	Indicator Arm	Carbon Steel	(2)	(2)
108	Spring Lockwasher	Carbon Steel, Zinc Plate	(2)	(2)
109	Split Clamp	Carbon Steel	(2)	(2)
110	Hexagon Bolt	Carbon Steel, Zinc Plate		

1. Alternative Materials
2. Same as Standard
3. Refer to Alternative Temperature Range under Specifications on page 3 for full range details.

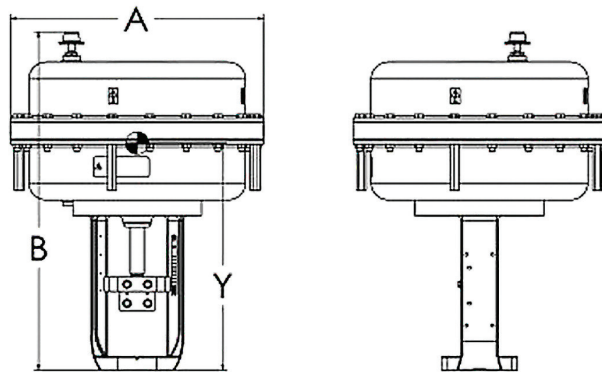
Dimensions and Weights (U.S.)



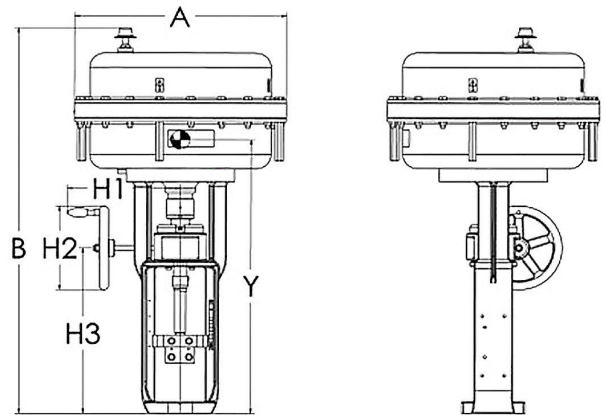
Shown without optional Handwheel



Shown with optional Handwheel



23L Shown without optional Handwheel



23L Shown with optional Handwheel

Dimensions and Weights

Actuator Size	Spring Range	Actuator Dimensions (inches)				H1	H2	H3	Weights (lbs.)	
		A	B	C	D				STD	W/HANDWHEEL
6	All	11.89	15.55 (17.52)	10	9	-	-	-	44	59.5
10	All	14.68	19.56 (21.53)	10.90	12	-	-	-	86	105.8
16	All	18.74	28.22 (30.78)	14	18	-	-	-	209	244.7
23	All	21.61	30.70 (33.26)	16	18	-	-	-	264.5	319.6
23L ⁽¹⁾	3-15, 6-30	21.61	27.79 (30)	-	-	-	-	-	374.8	-
23L ⁽¹⁾	11-23, 21-45	21.61	30.74 (32.95)	-	-	-	-	-	507	-
23L ⁽²⁾	3-15, 6-30	21.61	38.54 (40.74)	-	-	11.49	11.02	17.44	-	416.6
23L ⁽²⁾	11-23	21.61	41.49 (43.70)	-	-	11.49	11.02	17.44	-	548.9
23L ⁽²⁾	21-45	26.81	56.57 (53.34)	-	-	-	-	-	-	612.9

Actuator Removal Clearance = 6 inches

1. Without handwheel

2. With handwheel

Center of Gravity (inches)

Without Handwheel

Size	X	Y
6	0.2	9.8
10	0.0	12.9
16	0.1	18.5
23	0.1	21.1
23L ⁽³⁾	0.0	20.1
23L ⁽⁴⁾	0.0	21.9

3. Nominal spring ranges 3-15 & 6-30

4. Nominal spring ranges 11-23 & 21-45

With Handwheel

Size	XX	YY
6	1.3	9.1
10	0.9	12.0
16	1.4	16.7
23	1.4	19.0
23L ⁽³⁾	0.0	28.9
23L ⁽⁴⁾	0.0	30.7

Dimensions and Weights (U.S.)

Limit Stops (inches)

Up Stop

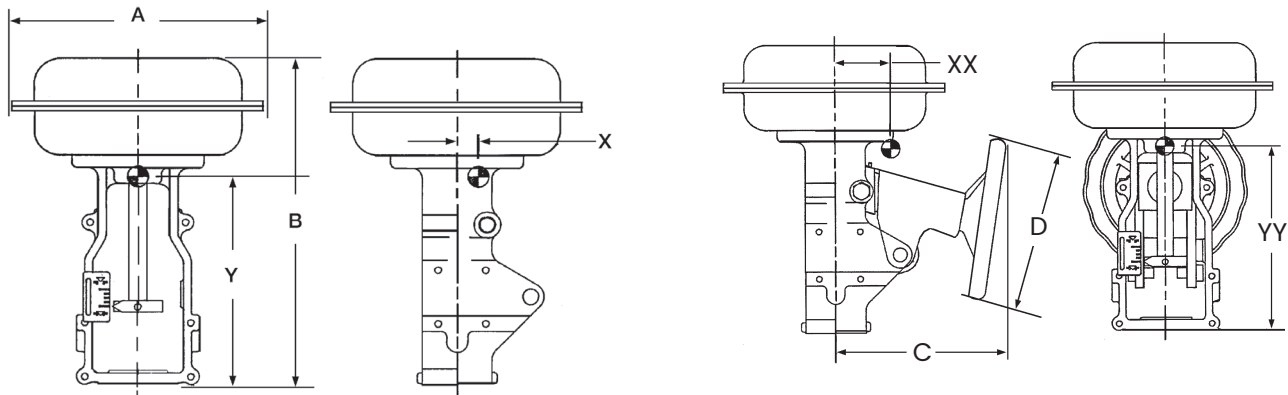
Size	Spring Range	Model	Overall Height B
6	All	87	19.5
10	All		25.4
16	All		36.4
23	All		38.8
23L ⁽¹⁾	3-15, 6-30		38.4
23L ⁽¹⁾	11-23, 21-45		41.3
23L ⁽²⁾	3-15, 6-30		49.1
23L ⁽²⁾	11-23, 21-45		52.1
6	All	88	19.2
10	All		25.1
16	All		35.5
23	All		35.5
23L ⁽¹⁾	3-15, 6-30		38.4
23L ⁽¹⁾	11-23, 21-45		41.3
23L ⁽²⁾	3-15, 6-30		49.1
23L ⁽²⁾	11-23, 21-45		52.1

1. Without Handwheel
2. With Handwheel

Down Stop

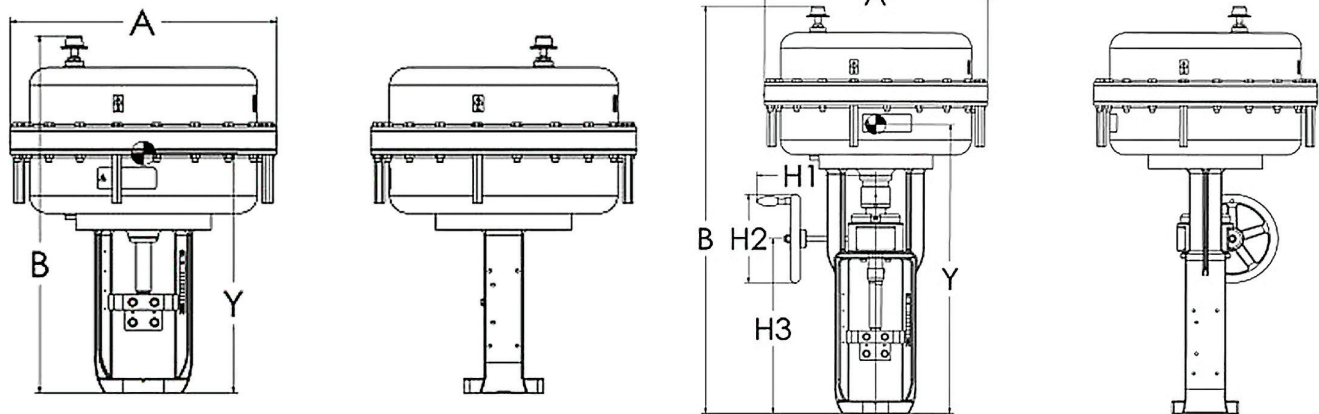
Size	Spring Range	Model	Overall Height B
6	All	87	19.8
10	All		26.0
16	All		37.2
23	All		39.9
23L ⁽¹⁾	3-15, 6-30		39.9
23L ⁽¹⁾	11-23, 21-45		42.8
23L ⁽²⁾	3-15, 6-30		50.6
23L ⁽²⁾	11-23, 21-45		53.6
6	All	88	19.7
10	All		25.9
16	All		37.5
23	All		40.3
23L ⁽¹⁾	3-15, 6-30		39.9
23L ⁽¹⁾	11-23, 21-45		42.8
23L ⁽²⁾	3-15, 6-30		50.6
23L ⁽²⁾	11-23, 21-45		53.6

Dimensions and Weights (Metric)



Shown without optional Handwheel

Shown with optional Handwheel



23L Shown without optional Handwheel

23L Shown with optional Handwheel

Dimensions and Weights

Actuator Size	Spring Range	Actuator Dimensions (mm)				H1	H2	H3	Weights (kg)	
		A	B	C	D				STD	W/HANDWHEEL
6	All	302	395 (445)	254	229	-	-	-	20	27
10	All	373	497 (547)	277	305	-	-	-	39	48
16	All	476	717 (782)	356	457	-	-	-	95	111
23	All	549	780 (845)	406	457	-	-	-	120	145
23L ⁽¹⁾	3-15, 6-30	549	706 (762)	-	-	-	-	-	170	-
23L ⁽¹⁾	11-23, 21-45	549	781 (837)	-	-	-	-	-	230	-
23L ⁽²⁾	3-15, 6-30	549	979 (1035)	-	-	292	280	443	-	189
23L ⁽²⁾	11-23	549	1054 (1110)	-	-	292	280	443	-	249
23L ⁽²⁾	21-45	681	1437 (1355)	-	-	-	-	-	-	278

Actuator Removal Clearance = 6 inches

1. Without handwheel
2. With handwheel

Center of Gravity (mm)

Without Handwheel

Size	X	Y
6	5	248
10	0	327
16	3	470
23	2	537
23L ⁽³⁾	0	511
23L ⁽⁴⁾	0	557

With Handwheel

Size	XX	YY
6	32	232
10	22	305
16	35	425
23	35	483
23L ⁽³⁾	0	734
23L ⁽⁴⁾	0	780

3. Nominal spring ranges 3-15 & 6-30
4. Nominal spring ranges 11-23 & 21-45

Dimensions and Weights (Metric)

Limit Stops (mm)

Up Stop

Size	Spring Range	Model	Overall Height B
6	All	87	494
10	All		646
16	All		925
23	All		987
23L ⁽¹⁾	3-15, 6-30		975
23L ⁽¹⁾	11-23, 21-45		1050
23L ⁽²⁾	3-15, 6-30		1248
23L ⁽²⁾	11-23, 21-45		1323
6	All	88	487
10	All		636
16	All		901
23	All		982
23L ⁽¹⁾	3-15, 6-30		975
23L ⁽¹⁾	11-23, 21-45		1050
23L ⁽²⁾	3-15, 6-30		1248
23L ⁽²⁾	11-23, 21-45		1323

1. Without Handwheel
2. With Handwheel

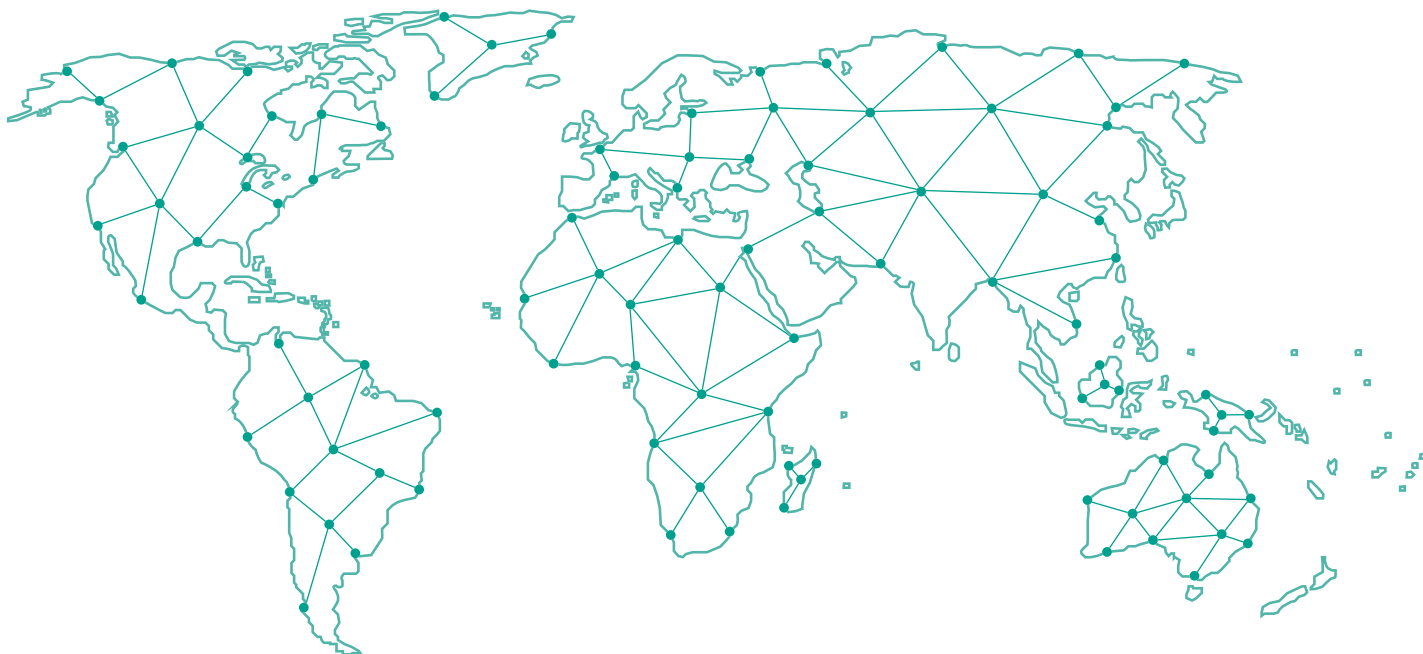
Down Stop

Size	Spring Range	Model	Overall Height B
6	All	87	503
10	All		660
16	All		945
23	All		1014
23L ⁽¹⁾	3-15, 6-30		1013
23L ⁽¹⁾	11-23, 21-45		1088
23L ⁽²⁾	3-15, 6-30		1286
23L ⁽²⁾	11-23, 21-45		1361
6	All	88	501
10	All		657
16	All		952
23	All		1024
23L ⁽¹⁾	3-15, 6-30		1013
23L ⁽¹⁾	11-23, 21-45		1088
23L ⁽²⁾	3-15, 6-30		1286
23L ⁽²⁾	11-23, 21-45		1361

Notes:

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